

Swiss Institute of Banking and Finance



University of St.Gallen

Case Study Session 5: Valuing Environmental, Social, and Governance (ESG): Financing Constraints and Workplace Safety

3,130/4,130 Corporate Finance

Key messages of this case study

1. You will learn the basic idea of **ESG**, i.e., environmental, social and governance factors, in corporate finance.
2. You will discuss an empirical research paper related to ESG.
 - This paper looks at the relationship between financing constraints and workplace safety.
 - You will learn how to cope with endogeneity in corporate finance and how to focus on causality rather than correlation.
 - You will get an introduction on how to structure and write an empirical paper in financial economics which is particularly important for students intending to write a bachelor thesis in finance.

ESG

1. Recap: What is ESG?
2. How is ESG measured?
3. Why is ESG important in corporate finance?

Recap: What is ESG?

ESG aims to incorporate environmental, social, and governance (ESG) factors in the investment decision of investors so that such investments facilitate firms' effort to run their businesses sustainably.

Environmental

- climate change – including physical risk and transition risk
- resource depletion, including water
- waste and pollution
- deforestation

Social

- working conditions, including slavery and child labor
- local communities, including indigenous communities
- conflicts
- health and safety
- employee relations and diversity

Governance

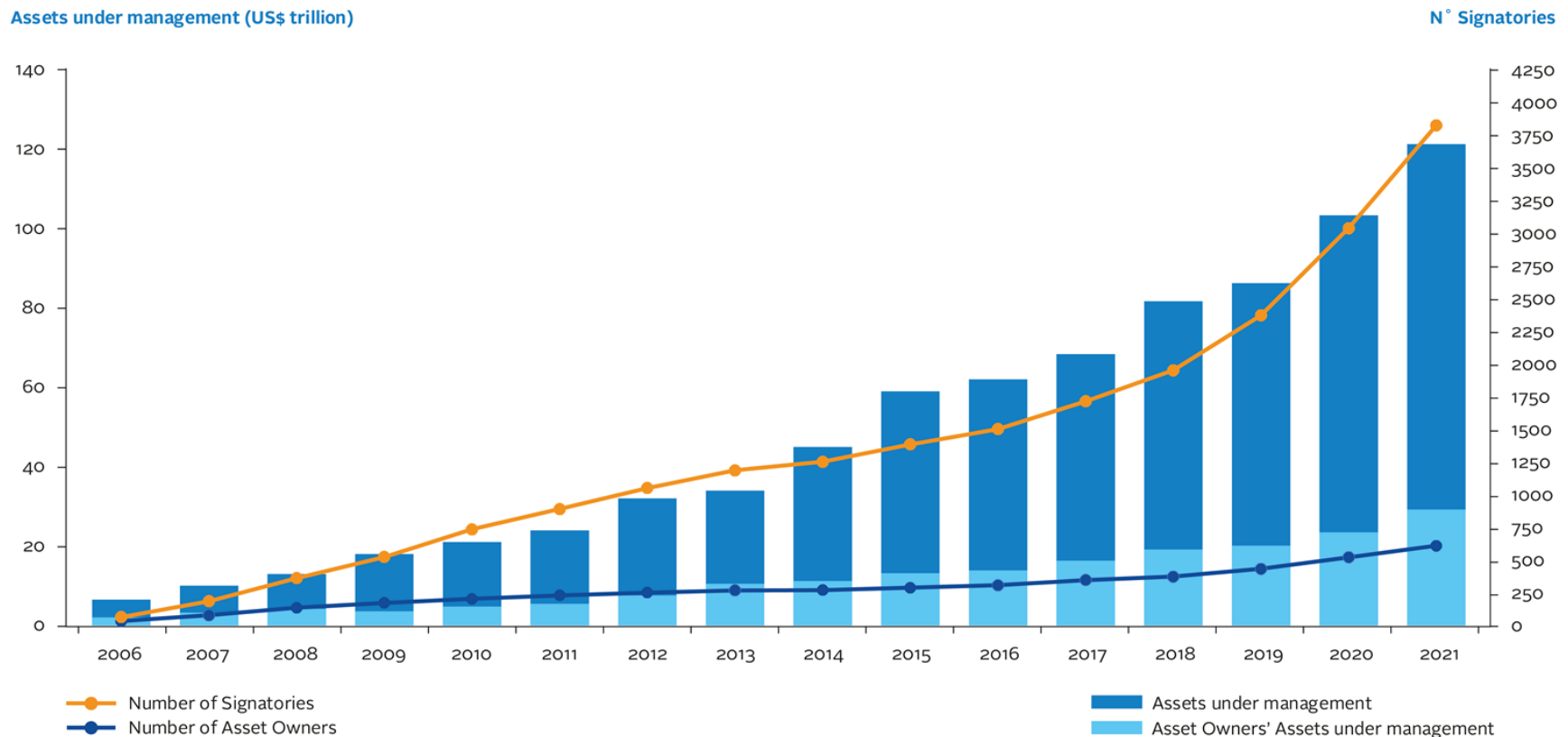
- executive pay
- bribery and corruption
- political lobbying and donations
- board diversity and structure
- tax strategy

Principles for Responsible Investment

<https://www.unpri.org/pri/what-is-responsible-investment>

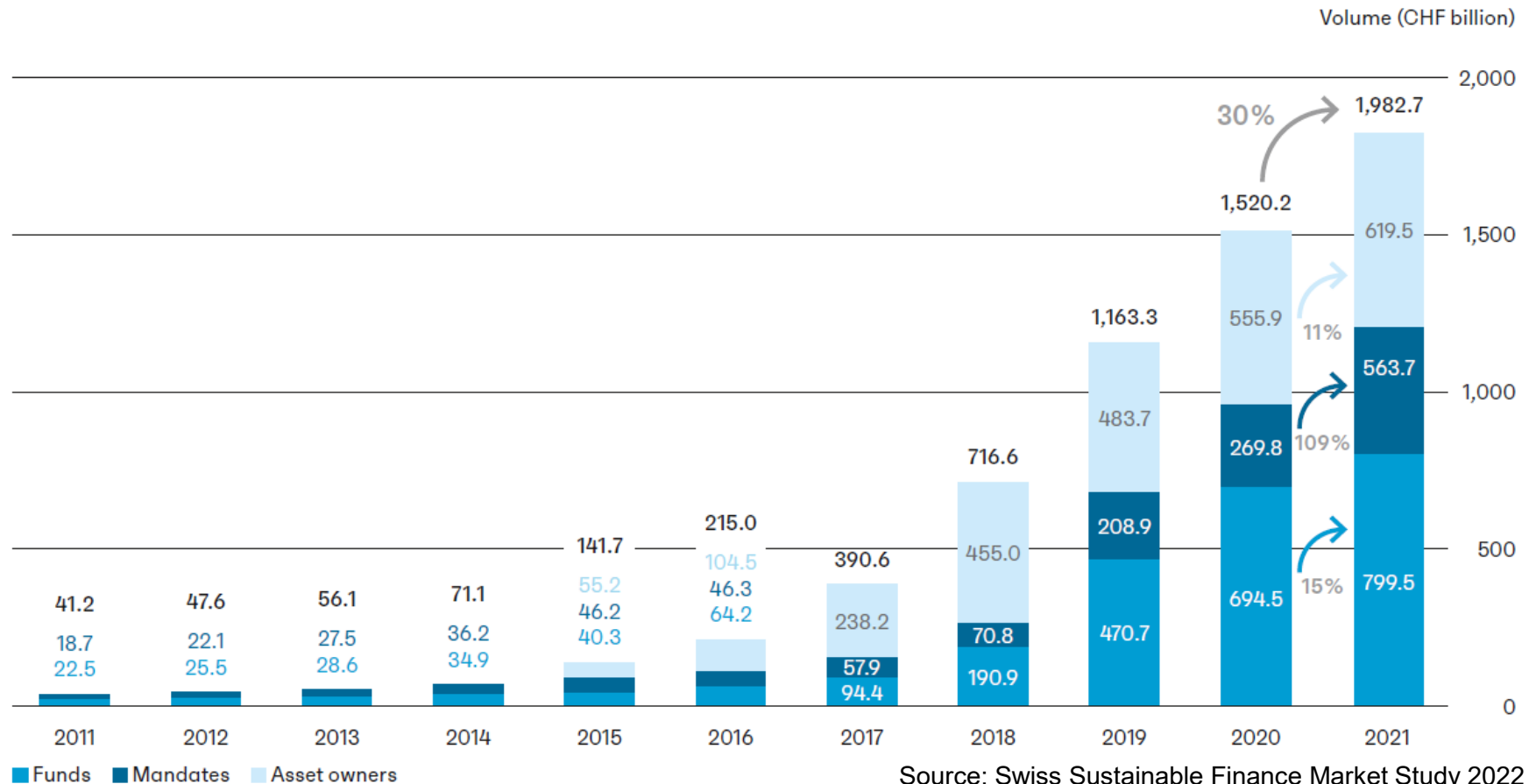
Development of ESG investments

- As of June 2023: 5,372 signatories; USD > 120 trillion AuM

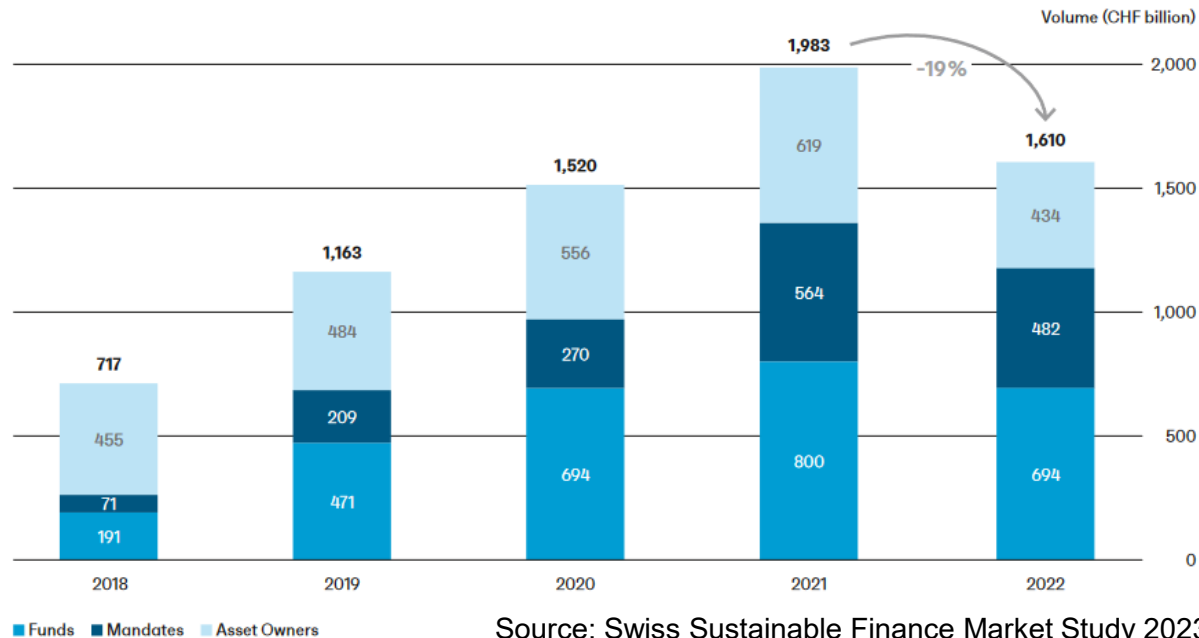


Source: Principles for Responsible Investment, Annual Report 2021

Development of ESG investments in Switzerland



Development of ESG investments in Switzerland



Main drivers of reported decline:

- The main driver (18%) is negative market performance in 2022.
 - Some market players have tightened their definitions of sustainability-related investments managed in Switzerland.
 - A new method for asset owners was implemented which led to lower reported volumes for stewardship approaches.
- ⇒ Net, there were still inflows.

How is ESG measured?

Negative Approach

Exclude firms engaged in controversial business from the investment process

→ e.g., weapons manufacturers, nuclear powerplants, etc.

Positive Approach

Absolute ratings
→ (binary) indicators

Within class ratings
→ (binary) indicators compared with peers (e.g., within industry)

Ratings

- Aggregation of the individual measures associated with the positive approach into a single rating metric
- Usually a number of sub-categories are defined (e.g., human rights, employee relations, diversity, etc.) and these sub-category-ratings are then aggregated into a total rating
- Ratings generally comprise strengths AND concerns of firms.



Who provides ESG data?

MSCI ESG Stats

(formerly KLD Stats)

*1988; USA; discontinued
(acq. in 2009 by RiskMetrics, in
2010 by MSCI)

Inrate

*1990 (Centre Info); *1995 (INrate),
Switzerland

MSCI IVA

(formerly Innovest)

*1992; USA
(acquired in 2009 by RiskMetrics
and in 2010 by MSCI)

Sustainalytics

(since 2019 Morningstar)

*1992; Netherlands

RepRisk

*1998, Switzerland

S&P Global

(formerly RobecoSAM)

*1999, Switzerland

FTSE4Good

*2001; UK

Refinitiv

(formerly Thomson Reuters &
Asset4), USA-UK

*2003; Switzerland

Moody's

(formerly Vigeo Eiris)

*2002; France
(acq. by Moody's in 2019)

ISS ESG Governance

QualityScore

*2013; USA

Truevalue Labs

*2013; USA

(since 2020 owned by FactSet)

Bloomberg ESG

*USA



Problem: Rating divergence

Correlations between ESG ratings at the aggregate rating level (ESG) and at the level of the environmental dimension (E), the social dimension (S), and the governance dimension (G) using the common sample. The results are similar using pairwise common samples based on the full sample. SA, SP, MO, RE, KL, and MS are short for Sustainalytics, S&P Global, Moody's ESG, Refinitiv, KLD, and MSCI, respectively.

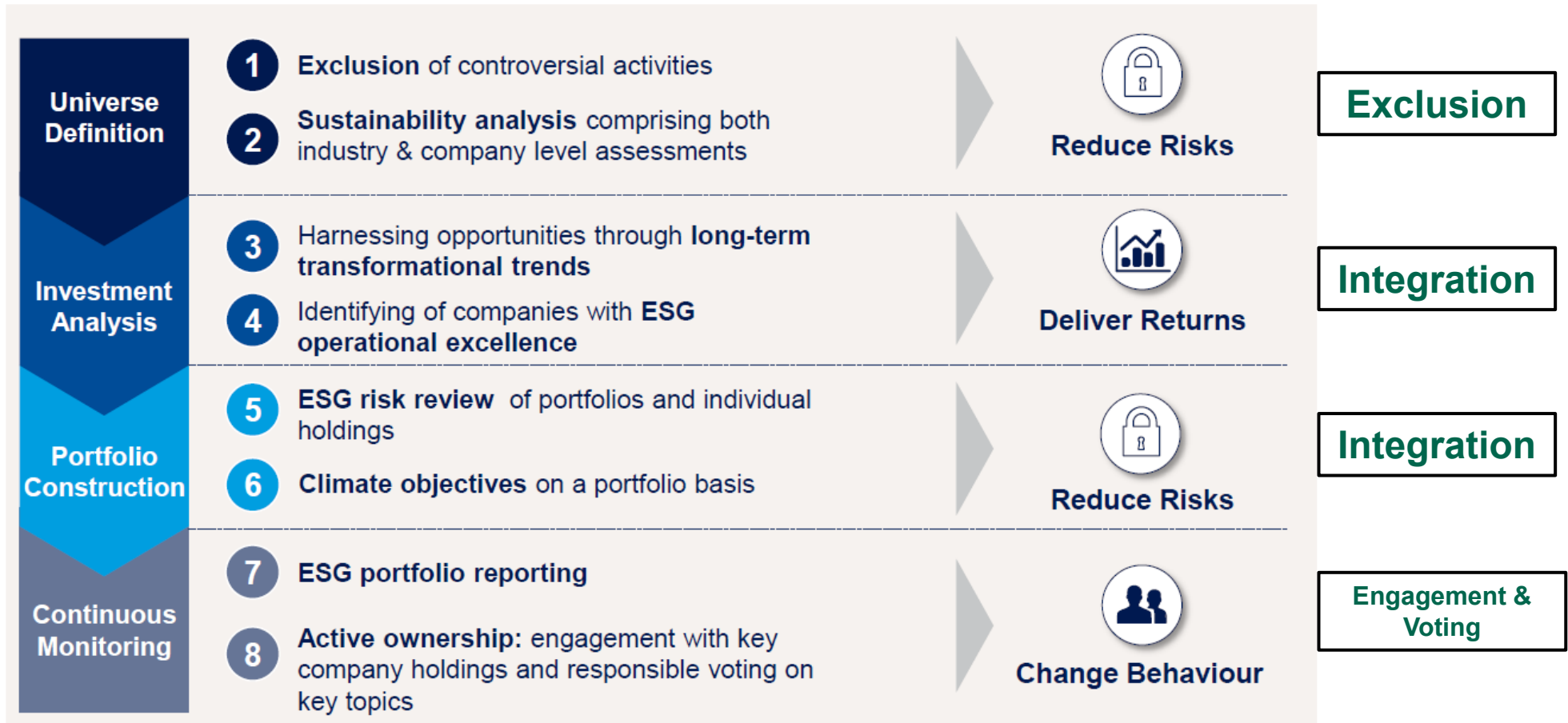
	KL SA	KL MO	KL SP	KL RE	KL MS	SA MO	SA SP	SA RE	SA MS	MO SP	MO RE	MO MS	SP RE	SP MS	RE MS	Average
ESG	0.53	0.49	0.44	0.42	0.53	0.71	0.67	0.67	0.46	0.7	0.69	0.42	0.62	0.38	0.38	0.54
E	0.59	0.55	0.54	0.54	0.37	0.68	0.66	0.64	0.37	0.73	0.66	0.35	0.7	0.29	0.23	0.53
S	0.31	0.33	0.21	0.22	0.41	0.58	0.55	0.55	0.27	0.68	0.66	0.28	0.65	0.26	0.27	0.42
G	0.02	0.01	-0.01	-0.05	0.16	0.54	0.51	0.49	0.16	0.76	0.76	0.14	0.79	0.11	0.07	0.30

Definitions of sustainable investment approaches

Best-in-Class	Approach in which a company's or issuer's ESG performance is compared with that of its peers based on a sustainability rating. All companies or issuers with a rating above a defined threshold are considered as investable.
ESG Engagement	Activity performed by shareholders with the goal of convincing management to take account of ESG criteria so as to improve ESG performance and reduce risks.
ESG Integration	The explicit inclusion by investors of ESG risks and opportunities into traditional financial analysis and investment decisions based on a systematic process and appropriate research sources.
ESG Voting	This refers to investors addressing concerns of ESG issues by actively exercising their voting rights based on ESG principles or an ESG policy.
Exclusions	An approach excluding companies, countries or other issuers based on activities considered not investable. Exclusion criteria (based on norms and values) can refer to product categories (e.g. weapons, tobacco), activities (e.g. animal testing), or business practices (e.g. severe violation of human rights, corruption).
Impact Investing	Investments intended to generate a measurable, beneficial social and environmental impact alongside a financial return. Impact investments can be made in both emerging and developed markets and target a range of returns from below-market to above-market rates, depending upon the circumstances.
Norms-Based Screening	Screening of investments against minimum standards of business practice based on national or international standards and norms.
Sustainable Thematic Investments	Investment in businesses contributing to sustainable solutions, both in environmental or social topics.

Source: Swiss Sustainable Finance Market Study 2021

How can SI strategies be implemented? – An example



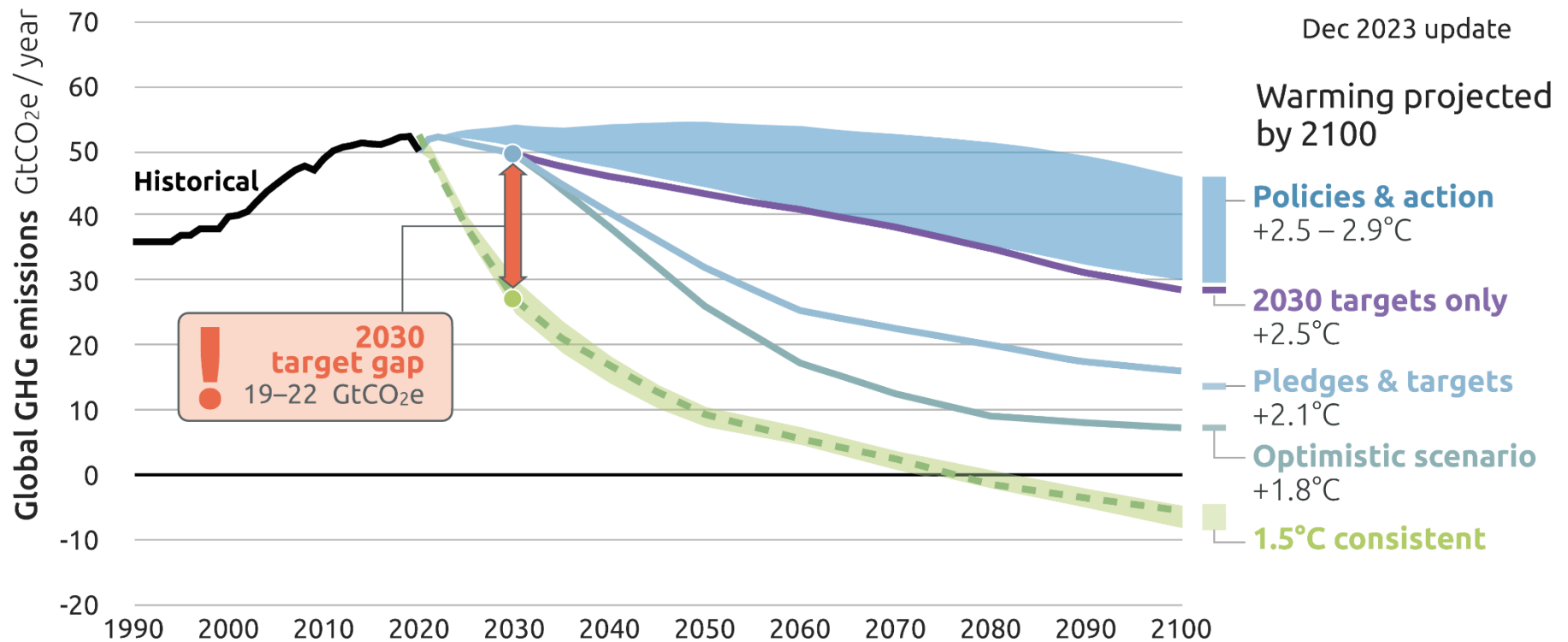
Why is ESG important in corporate finance?

Some empirical evidence

- Demand for ESG investments
- ESG and trust in corporations
- ESG and institutional investors
- Environmental externalities and cost of capital

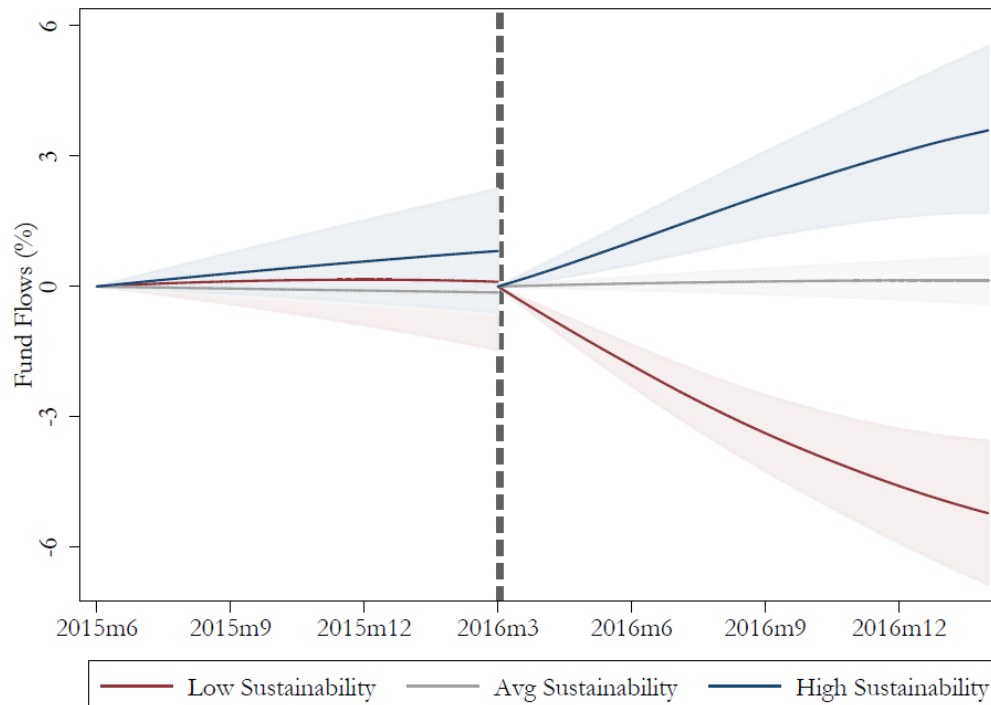
Climate change as main driver of demand for ESG investments

- Warming scenarios by Climate Action Tracker:



Demand for ESG investments

Fund Flows after Sustainability Rating



Source: Hartzmark and Sussman (2019)

- Hartzmark & Sussman (2019) show that investors prefer funds with high sustainability ratings.
- In total, investors withdrew more than \$12 billion from low sustainability funds and invested more than \$24 billion into highly rated funds.
- The authors look at the point in time when Morningstar first published its sustainability ratings.
- However, there is no evidence that high sustainability funds outperform low sustainability funds.

ESG and trust in corporations

- Lins, Servaes, and Tamayo (2017) investigate the impact of corporate social responsibility (CSR) intensity on firm performance during the 2008-2009 financial crisis.
- They focus on the channel that (investment in) social capital can maintain and increase trust in investment firms and hence impact firm performance.
- They explore the financial crisis as exogenous variation in the trust in firms.
- Firms with high CSR (ESG) ratings experienced significantly higher (between four and seven percentage points) stock returns during the financial crisis than those with low CSR ratings. Hence, investors assign a premium to high-CSR firms.
- High-CSR firms had higher profitability and gross margins, and experienced higher sales growth. They were also able to raise more debt during the crisis. These results are consistent with stakeholders and investors considering high-CSR firms to be more trustworthy during the crisis (a time of generally low trust).

ESG and institutional investors

- Dyck, Lins, Roth, and Wagner (2019) assess whether shareholders drive the environmental and social (E&S) performance of firms worldwide.
- Institutional ownership is positively associated with E&S performance, i.e., ESG ratings.
- The effect is even larger after shocks that reveal financial benefits to E&S improvements (e.g., BP Deepwater Horizon oil spill). Firms with higher institutional ownership at the time of the shock are more reactive in improving environmental performance in the following years than firms with low institutional ownership.
- Institutional investors convey their preferences for improved E&S by privately engaging with the firms they already own.
- Foreign institutional investors impact firms' E&S performance only when these investors are from countries with higher demand for E&S performance.

Environmental externalities and cost of capital

- Chava (2014) analyzes the impact of a firm's environmental profile on its cost of (equity and debt) capital.
- Investors demand significantly higher expected returns on stocks excluded by environmental screens compared to firms without such environmental concerns.
- Lenders also charge a significantly higher interest rate on the bank loans issued to firms with environmental concerns.
- Firms with environmental concerns have lower institutional ownership, and fewer banks participate in their loan syndicate than firms without environmental concerns.
- These results suggest that exclusionary socially responsible investing and environmentally sensitive lending can have a material impact on the cost of (equity and debt) capital of affected firms.

Research Paper:

Financing constraints and workplace safety

1. What is the research question of the paper?
2. How is the paper motivated?
3. How is the paper related to our course?
4. What is the relationship between investment and financing constraints?
5. What is the result of the main regression framework?
6. What are potential difficulties in the main regression framework? What are potential solutions to address these problems?
7. What are the authors' strategies in the paper to address these problems?
8. What are the results of these strategies?
9. What do we learn?

Financing constraints and workplace safety

Research question

What is the research question?

Financing constraints and workplace safety

Motivation of the paper

How is this paper motivated? Why is it relevant?

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Relation to the course

How is the paper related to our course?

Financing constraints and workplace safety

Financing constraints

What are financing constraints?

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Economic intuition

What is the relationship between investment and financing?

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Investments

What are typical investments to improve workplace safety?

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Costs & benefits

What are costs and benefits for firms to invest in workplace safety?

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Dependent variables

What is the main dependent variable?

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Dependent variables

Are injury rates simply explained by (dangerous) industries?

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Main empirical results

Please explain the explanatory variables in the main regression specification?

$$\frac{Injuries}{Hour_{ij}} = \alpha_i + \delta_{jt} + \gamma_{st} + \beta X_{it} + \epsilon_{it}$$

Financing constraints and workplace safety

Main empirical results

Model Dependant variable Sample	OLS Inj/Hour All (1)	OLS Inj/Hour All (2)	OLS Inj/Hour All (3)
Debt/Assets	0.0047** (0.0024)	0.0075** (0.0036)	0.0063** (0.0029)
Cash/Assets	-0.0103** (0.0050)	-0.0171** (0.0068)	-0.0016 (0.0031)
CashFlow/Assets	-0.0051* (0.0029)	-0.0033 (0.0038)	0.0023 (0.0033)
Dividends/Assets	-0.0139* (0.0079)	-0.0098 (0.0121)	-0.0218** (0.0088)
Log(Assets)	-0.0020*** (0.0006)	-0.0010** (0.0005)	-0.0040*** (0.0012)
AssetTurnover	0.0033*** (0.0012)	0.0052*** (0.0015)	0.0005 (0.0010)
MarketToBook	-0.0003 (0.0006)	-0.0004 (0.0007)	0.0002 (0.0004)
TangibleAssetRatio	0.0267*** (0.0072)	0.0376*** (0.0090)	0.0106* (0.0062)
Capex/Assets	-0.0397** (0.0160)	-0.0576** (0.0234)	0.0052 (0.0080)
Log(Employees)	0.0007 (0.0004)	0.0053*** (0.0017)	-0.0006 (0.0009)
Hours/Employee	-0.0000*** (0.0000)	-0.0000 (0.0000)	-0.0000*** (0.0000)
Establishment fixed effects	No	Yes	Yes
Year × Industry dummies	No	No	Yes
Year × State dummies	No	No	Yes
Observations	43,371	43,371	25,053
Adjusted R^2	0.0273	0.0490	0.0618

1. What do the coefficients on the first three variables tell us?
2. What is the economic effect of the coefficient on Debt/Assets?
3. Why are there fewer observations in specification (3) than in (1) and (2)?

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Main empirical results

What do the coefficients on the first three variables tell us?

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Main empirical results

What is the economic effect of the coefficient on Debt/Assets?

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Main empirical results

Why are there fewer observations in specification (3)?

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Endogeneity problems

Do you see any endogeneity problems associated with the previous regression framework?

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Endogeneity problems

How do the authors address these endogeneity problems?

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Quasi-natural experiments

What is the basic idea of a quasi-natural experiment?

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Quasi-natural experiments

How would a quasi-natural experiment estimated with a difference-in-differences estimator look like vs. the OLS spec.?

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Quasi-natural experiment: AJCA

How does the American Job Creation Act work as a quasi-natural experiment?

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Quasi-natural experiment: Debt maturity structure

How can the maturity structure of firms' debt at the onset of the financial crisis be used as a quasi-natural experiment?

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Quasi-natural experiment: Oil price fluctuations

How can changes in the oil price be used as a quasi-natural experiment?

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Quasi-natural experiment: AJCA

How do the quasi-natural experiments work empirically?
Let's look at the example of the American Job Creation Act!

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Quasi-natural experiment

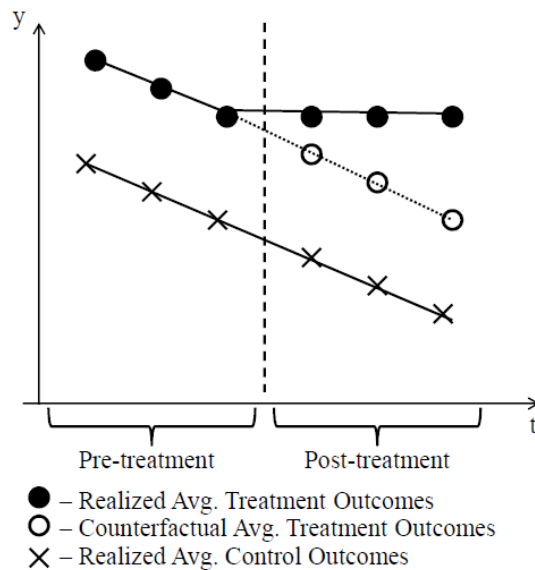
Do the quasi-natural experiments support the OLS results?

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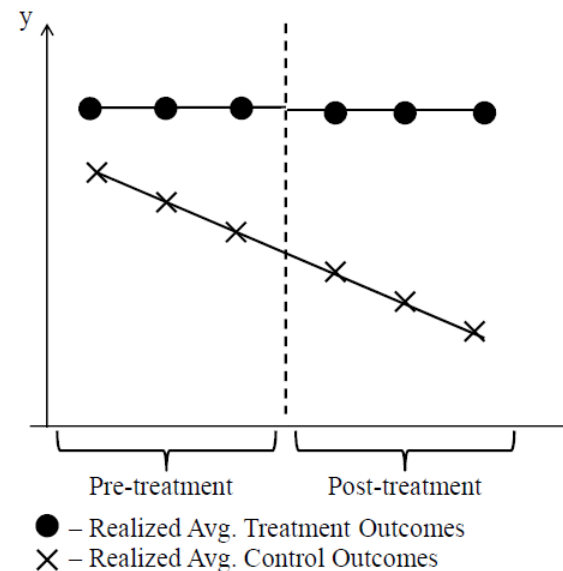
Quasi-natural experiment: Parallel trends

Differences-in-differences analyses heavily rely on the “common trend” assumption, i.e., the dependent variables move in parallel.

Common Trend



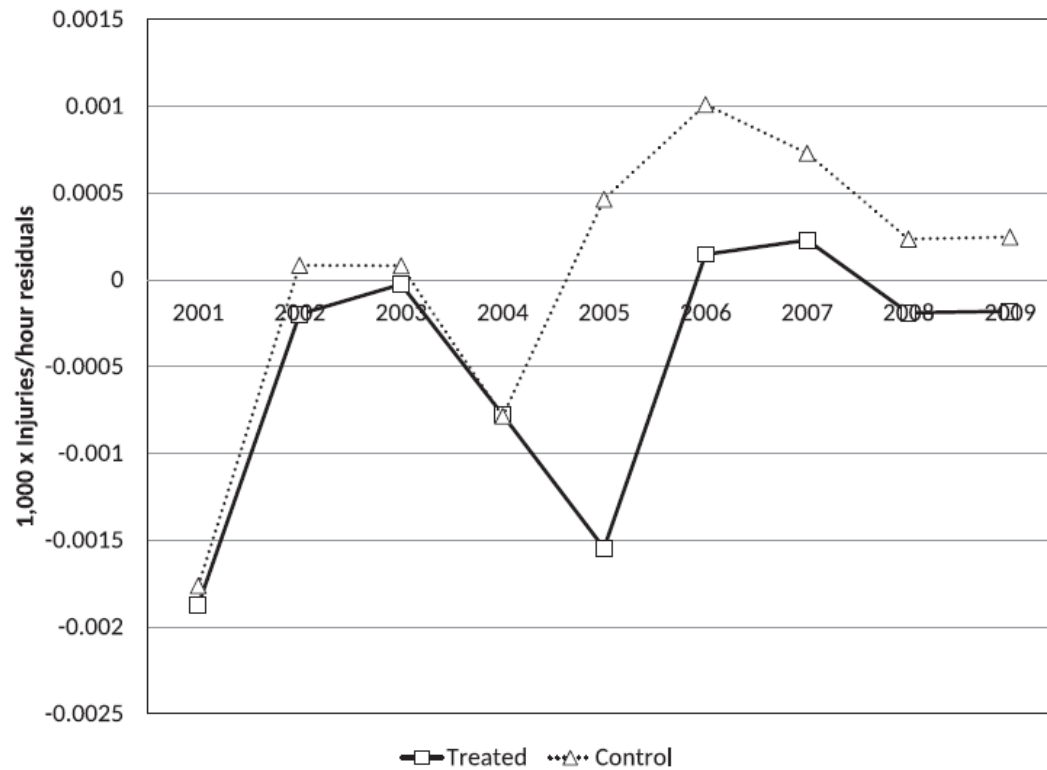
No Common Trend



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Quasi-natural experiment: Parallel trends

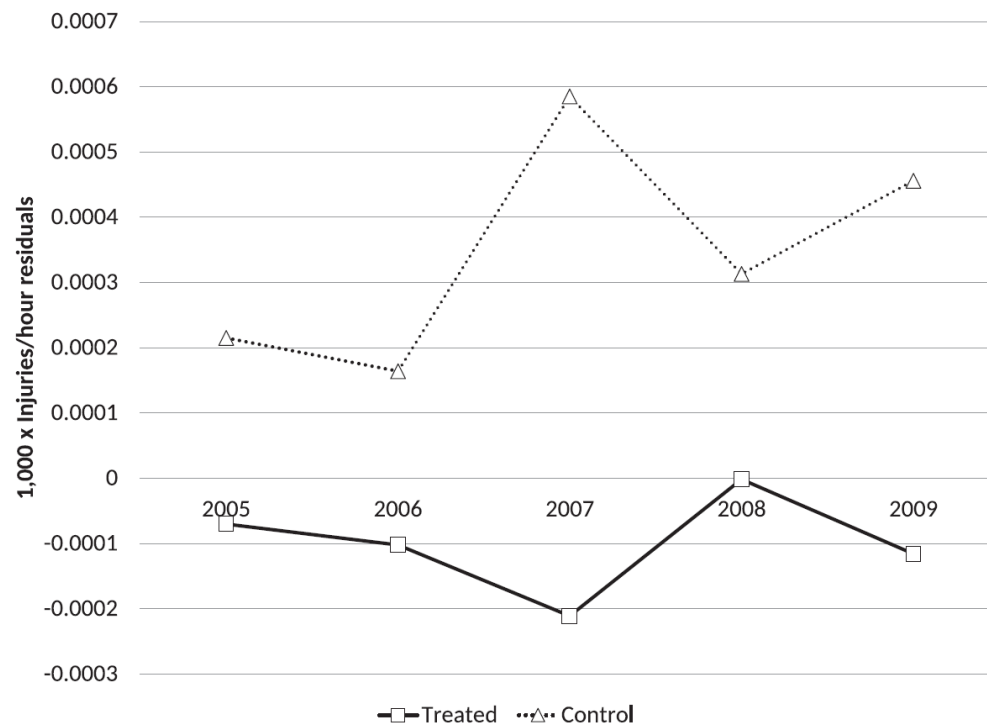
Does the graphical inspection of AJCA support parallel trends?



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Quasi-natural experiment: Parallel trends

Does the graphical inspection of the maturity structure of debt at the onset of the financial crisis support results?



Financing constraints and workplace safety

Firm value and injury rates

Do firms have to bear any costs from higher injury rates?

Recap

- This paper provides new evidence that injury rates increase with leverage and decrease with operating cash flow.
- Financing can affect the well-being of employees through limited investments in workplace safety.
- The results also have implications for optimal financial policy, as firm value appears to decline with injury rates.

Literature

Cohn, J.B., and M.I. Wardlaw, 2016, Financing constraints and workplace safety, *Journal of Finance* 71, 2017-2058.

Additional literature

- Hartzmark, S.M., and A.B. Sussman, 2019, Do investors value sustainability? A natural experiment examining ranking and fund flows, *Journal of Finance*, forthcoming .
- Lins, K.V., Servaes, H., and A. Tamayo, 2017, Social capital, trust, and firm performance: The value of corporate social responsibility during the financial crisis, *Journal of Finance* 72, 1785-1824.
- Dyck, A., Lins, K.V., Roth, L., and H.F. Wagner, 2019, Do institutional investors drive corporate social responsibility? International evidence, *Journal of Financial Economics* 131, 693-714.
- Chava, S., 2014, Environmental externalities and cost of capital, *Management Science* 60, 2223-2247.
- Roberts, M.R., and T.M. Whited, 2013, Endogeneity in empirical corporate finance, In: *Handbook of the Economics of Finance* 2, 493-572.